R15

Code No: 124AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2018
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define DFA. [2]
b) Design FA which accepts set of all strings ending with 00. [3]
c) Define Left linear Grammar. [2]
d) Give the regular expression for the language all string over alphabet **{0,1}** containing at least two 0's. [3]
e) What is ambiguity in CFG? [2]
f) Write the context free grammar for the language **$L=\{a^n b^{2n}/n\geq 1\}$** [3]
g) Give Instantaneous description ID of Turing Machine. [2]
h) Define Type 0 Grammar. [3]
i) List any 2 NP Hard Problems. [2]
j) Define Turing reducibility. [3]

PART-B

(50 Marks)

2.a) Convert the following NFA with € - moves to DFA shown in figure 1.

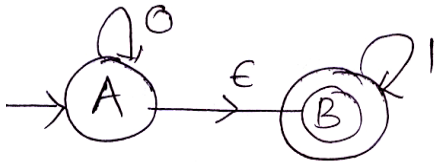


Figure: 1

b) Minimize the following DFA shown in figure 2. [5+5]

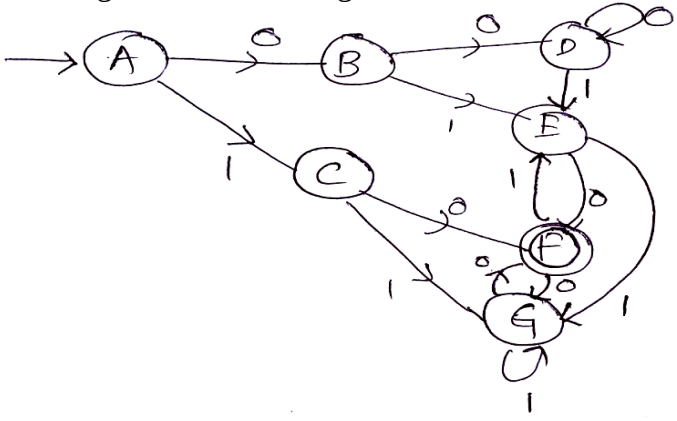


Figure: 2

OR

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3.a) Check whether the following two Finite Automaton's are equivalent or not?
Finite Automaton (FA) 1 (figure 3):



Figure: 3

Finite Automaton (FA) 2(figure 4):



Figure: 4

b) Convert the following NFA with € moves to DFA in figure 5. [5+5]

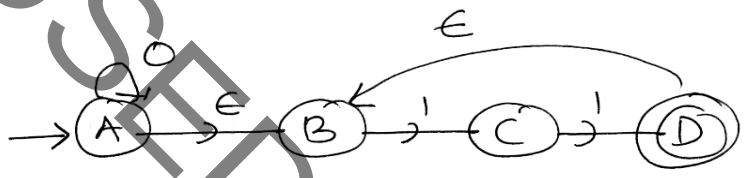


Figure: 5

4. Construct an NFA for the following Regular expression:
a) 01[((10)*+111)*+0]*1
b) ((01+10)*00)* [5+5]

OR

5.a) Find the regular grammar for the following Finite Automate shown in figure 6.

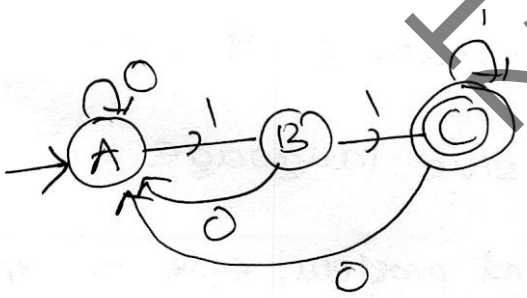


Figure: 6

b) Construct FA for the following regular expressions (0+1)*(1+00)(0+1)*. [5+5]

6.a) Convert the following grammar to Chomsky Normal Form
S→ ABA
A→ aA | Є
B→ bB | Є
and simplify the grammar

b) Write and explain closure properties of Context Free Languages. [5+5]

OR

7.a) State the Pumping Lemma for Context Free Languages.

b) Design Push down Automata for the language **L = {$a^n b^{2n}$ | n≥1}.** [5+5]

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8.a) Design Turing Machine for the Language L={ $\mathbf{a^n b^n c^n / n \geq 1}$ }

b) List the Closure properties of recursive Languages. [6+4]

OR

9.a) Design Turing Machine to compute the function n!

b) Design TM for performing proper subtraction of two numbers. [5+5]

10.a) Briefly write about Universal Turning Machine (UTM).

b) What do you mean by NP Complete? List any 6 NP Complete Problems. [4+6]

OR

11.a) Discuss about turing Reducibility.

b) Write about:

i) Post Correspondence Problem

ii) Halting problem of TM. [3+7]

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